```
        .build();    #B
var response = httpClient.send(request,    #C
        HttpResponse.BodyHandlers.ofString());    #C
```

#A Build the form content with the assertion JWT
#B Create the POST request to the token endpoint
#C Send the request and parse the response

Putting this all together, listing 11.3 shows a sample client application that loads the public and private key pair from the keystore, outputs the public key as a JWK Set, and then signs a client assertion and sends it to the AS. Create a new file named JwtBearerClient.java in the folder src/main/java/com/manning/apisecurityinaction and create a class named `JwtBearClient` inside it. Add an executable `main()` method with the contents of listing 11.3. You can compile and run the file to obtain an access token using JWT bearer authentication.

> **TIP** If the connection is refused due to SSL errors refer to section 7.4.2 of chapter 7 for details on how to configure Java's HTTP client to trust the server certificate.

Listing 11.3 JWT bearer authentication client

```
var password = "changeit".toCharArray();     #A
var keyStore = KeyStore.getInstance("PKCS12");    #A
keyStore.load(new FileInputStream("keystore.p12"),    #A
        password);    #A
var privateKey = (ECPrivateKey) keyStore.getKey("es256-
key",    #A
        password);    #A
```